

Solution Requirements

Date	1 July 2026
Team ID	SWTID-2026-1614
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Optional user sign-up and login for policy researchers/developers to save simulation histories and comparative tracks. Passwords are encrypted using bcrypt.	Low
2	Authorization levels	Two user levels are supported: Public Guest (can search countries and run simulations) and Research Admin (can modify baseline datasets and trigger model retraining).	Medium
3	External interfaces	The system connects to the serialized Linear Regression model (model.pkl) and a local SQLite database or flat file (HDI.csv) to pre-populate country baselines.	High
4	Transactions processing	Each prediction request containing target indicators (Life Expectancy, Schooling, GNI) is validated, processed by the ML engine, and logged in real-time.	High
5	Reporting	The system generates a clean policy simulation report displaying the predicted score, developmental tier, and dynamic gauge comparison charts.	Medium
6	Business rules, etc.	Computations are only triggered if values pass logic validation checks: Life Expectancy (20–100 years), Mean Schooling (0–25 years), and GNI (\$100–\$150,000).	High
7	Compliance to laws or regulations	The system stores no sensitive personal or financial information, minimizing regulatory issues. Data usage adheres to UNDP public-use terms.	Low
8	Other	The app supports both instant baseline country profiling and manual parameter overrides from the same glassmorphism form interface.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The prediction calculations and UI visual cards must load quickly after the user initiates the prediction check.	Result returned in under 1 second
2	Scalability	The application should handle multiple concurrent policy researchers simulating development plans without slowing down.	Supports up to 500 concurrent active users
3	Security & Data Privacy	Protection of the admin upload pipeline and securing API transport layers.	All communications routed over HTTPS; admin endpoints secured with JWT tokens
4	Reliability & Availability	The web application must remain highly accessible during school hours and academic presentation periods.	99.5% uptime per month
5	Usability & Accessibility	Clear, readable font hierarchy with modern glassmorphism styles designed to be easily navigated on mobile, tablet, and desktop viewports.	Fully mobile-responsive interface
6	Other	The project files must be maintainable, enabling easy replacement of the HDI.csv database or updating model weights with zero downtime.	Updates and redeployments completed within 30 minutes